USN. | | | | | | | | | | [24MTASD102]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY- 2025

STRUCTURAL DYNAMICS

Time: 3 Hours **MaximumMarks:100**

Instructions: **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.

ii. Missing data may be suitably assumed.

			COs	Blooms Level	Marks
1.	**a.**	Derive equation of motion for a single-degree-of-freedom spring-mass - damper system using Lagrange's Equation.	**1**	**3**	**10**
	b.	Examine the variation of magnification factor and phase angle of a single degree of freedom spring-mass-damper system due to harmonic excitation with damping ration and frequency ratio?	**1**	**4**	**10**
2.	**a.**	Derive equation of motion for a multi-degree-of-freedom spring-mass-damper system using Lagrange's Equation.	**2**	**3**	**10**
	b.	For a multi-degree-of-freedom undamped systems derive modal equation.	**2**	**3**	**10**
3.	**a.**	Derive governing differential equation for a vibrating string.	**3**	**3**	**6**
	b.	Find equation of motion of a cantilever beam using single assumed mode in conjunction with Rayleigh-Ritz method.	**3**	**3**	**14**
		OR			
4.	**a.**	What are the matrix properties for which the eigenvalues are zero, real or complex?	**4**	**2**	**10**
	b.	For a multi-degree-of-freedom systems, investigate the methods used for eigenvector normalization.	**4**	**4**	**10**
5.	**a.**	Examine different types of boundary conditions used for beam vibration analysis.	**5**	**4**	**10**
	b.	Distinguish between first three modes of vibration of a clamped-clamped beam and a simply supported beam.	**5**	**2**	**10**
		OR			
6.	**a.**	What is a transducer? Compare different types of transducers used for vibration measurements.	**5**	**2**	**10**
	b.	Analyze the expression for Frequency Response fFunction (FRF) for a single-degree-of-freedom spring-mass-damper system.	**5**	**4**	**10**

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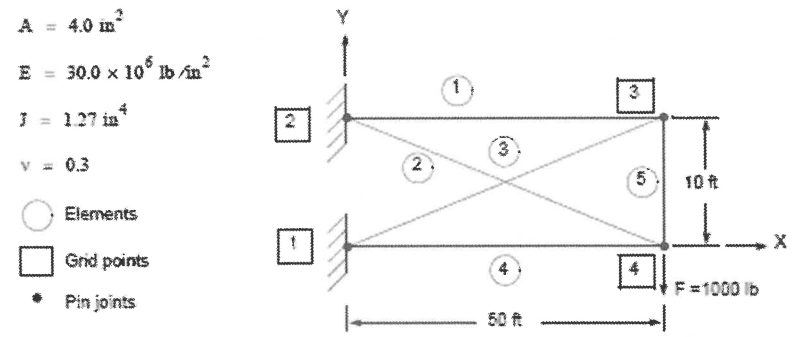
7.	**a.**	Illustrate the difference between geometric, material and contact nonlinearities? Provide typical situation where you encounter such nonlinearities.	**6**	**2**	**10**
	b.	Illustrate the response of a single degree-of-freedom linear spring-mass-damper system due to harmonic excitation. Superimpose on it the effect of spring hardening and spring softening.	**6**	**2**	**10**
		OR			
8.	**a.**	Why random vibration is important in Aeronautical Engineering? Give some examples where one experience random input to the Aeronautical Systems	**6**	**2**	**10**
	b.	Explain the steps followed to prepare Nastran input file to solve following truss problem.  Figure 2-1 Truss Example	**6**	**2**	**10**